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FASTNERS

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Fasteners

A machine element used for holding or joining two or more parts of a machine or structure is known as a fastener. The process of joining the parts is called fastening. The fasteners are of two types : permanent and removable (temporary). Riveting and welding processes are used for fastening permanently. Screwed fasteners such as bolts, studs and nuts in combination, machine screws, set screws, etc., and keys, cotters, couplings, etc., are used for fastening components that require frequent assembly and disassembly.

Screwed fasteners occupy the most prominent place among the removable fasteners. In general, screwed fasteners are used : (i) to hold parts together, (ii) to adjust parts with reference to each other and (iii) to transmit power.

5.2 SCREW THREAD NOMENCLATURE

A screw thread is obtained by cutting a continuous helical groove on a cylindrical surface (external thread). The threaded portion engages with a corresponding threaded hole (internal thread); forming a screwed fastener. Following are the terms that are associated with screw threads (Fig. 5.1).

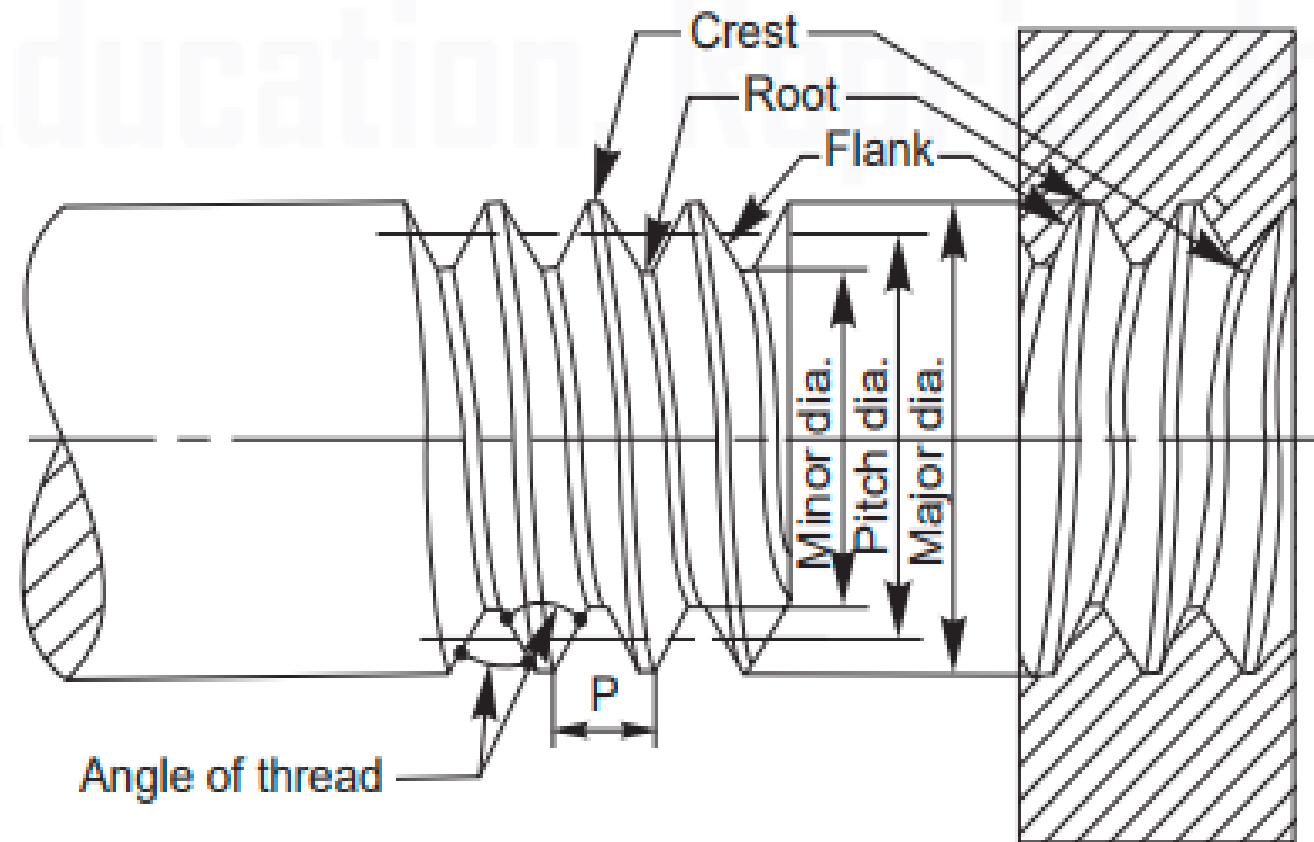


Fig. 5.1 Screw thread nomenclature

1. Major (nominal) diameter

This is the largest diameter of a screw thread, touching the crests on an external thread or the roots of an internal thread.

2. Minor (core) diameter

This is the smallest diameter of a screw thread, touching the roots or core of an external thread (root or core diameter) or the crests of an internal thread.

3. Pitch diameter

This is the diameter of an imaginary cylinder, passing through the threads at the points where the thread width is equal to the space between the threads.

4. Pitch

It is the distance measured parallel to the axis, between corresponding points on adjacent screw threads.

5. Lead

It is the distance a screw advances axially in one turn.

6. Flank

Flank is the straight portion of the surface, on either side of the screw thread.

7. Crest

It is the peak edge of a screw thread, that connects the adjacent flanks at the top.

8. Root

It is the bottom edge of the thread that connects the adjacent flanks at the bottom.

9. Thread angle

This is the angle included between the flanks of the thread, measured in an axial plane.

5.3 FORMS OF THREADS

Bureau of Indian Standards (BIS) adapts ISO (International Organisation for Standards) metric threads which are adapted by a number of countries apart from India.

The design profiles of external and internal threads are shown in Fig. 5.2. The following are the relations between the various parameters marked in the figure :

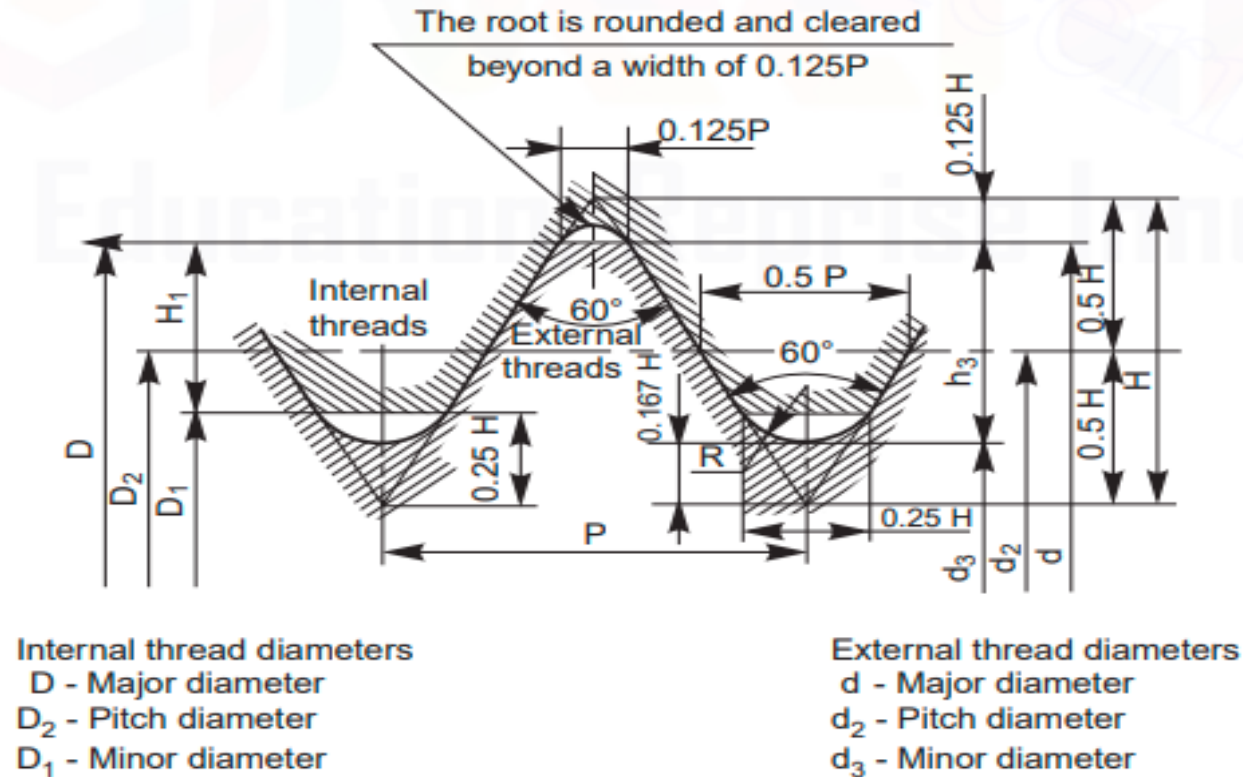


Fig. 5.2 Metric screw thread

$$\begin{aligned}P &= \text{Pitch} \\H &= 0.86 P \\D &= d = \text{Major diameter} \\D_2 &= d_2 = d - 0.75H \\D_1 &= d_2 - 2(H/2 - H/4) = d - 2H_1\end{aligned}$$

$$\begin{aligned}d_3 &= d_2 - 2(H/2 - H/6) \\&= d - 1.22P \\H_1 &= (D - D_1)/2 = 5H/8 = 0.54P \\h_3 &= (d - d_3)/2 = 17/24H = 0.61P \\R &= H/6 = 0.14P\end{aligned}$$

5.3.1 Other Thread Profiles

Apart from ISO metric screw thread profile, there are other profiles in use to meet various applications. These profiles are shown in Fig. 5.3, the characteristics and applications of which are discussed below :

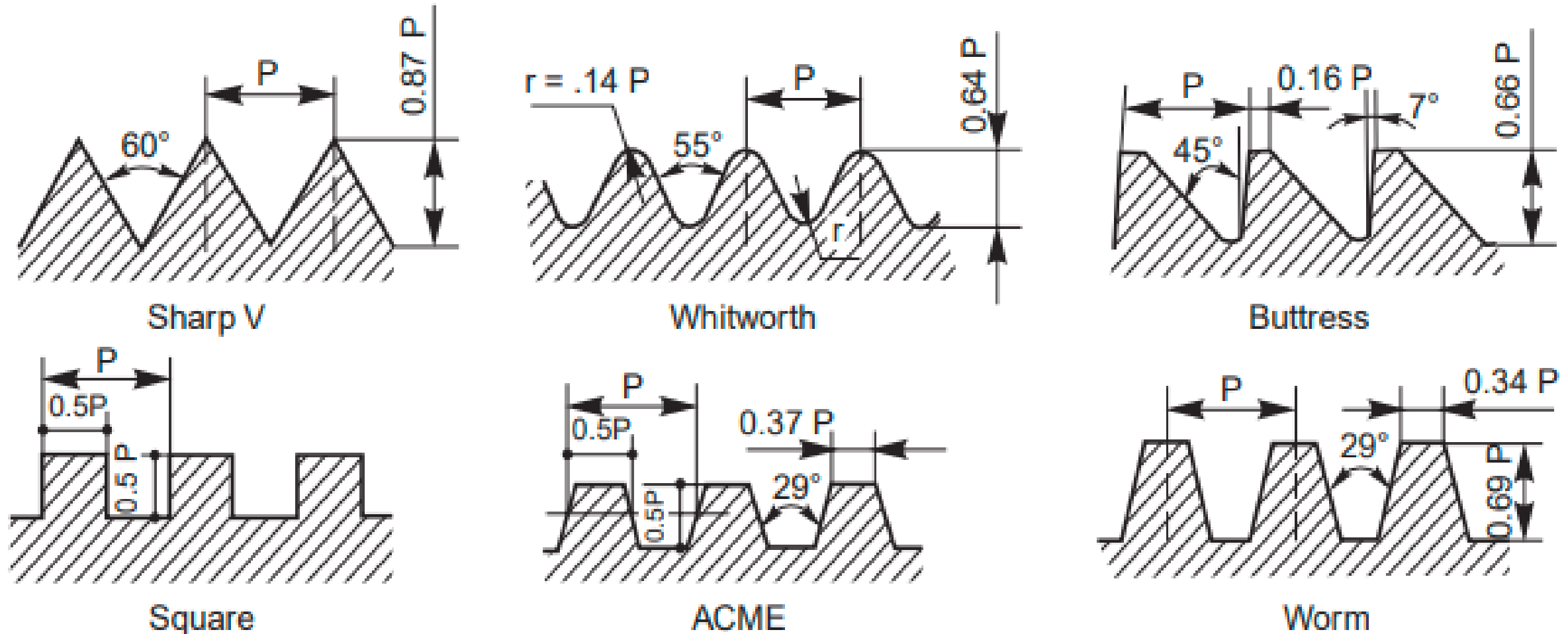


Fig. 5.3 Types of thread profiles

5.4 THREAD SERIES

BIS recommends two thread series: coarse series and fine series, based on the relative values of the pitches. However, it must be noted that the concept of quality is not associated with these terms. For any particular diameter, there is only one largest pitch, called the coarse pitch and the rest are designated as fine pitches.

Table 5.1 gives the nominal diameter and pitch combinations for coarse and fine series of ISO metric screw threads.

Table 5.1 Diameter-pitch combination for ISO metric threads

Nominal diameter		Pitch			
First choice	Second choice	Coarse	Fine		
			1	2	3
2	—	0.4	0.25	—	—
—	2.2	0.45	0.25	—	—
2.5	—	0.45	0.35	—	—
3	—	0.5	0.35	—	—
—	3.5	0.6	0.35	—	—
4	—	0.7	0.5	—	—
—	4.5	0.75	0.5	—	—
5	—	0.8	0.5	—	—
6	—	1	0.75	0.5	—
8	—	1.25	1	0.75	—
10	—	1.5	1.25	1	0.75
12	—	1.75	1.5	1.25	—
16	14	2	1.5	1	—
20	18,22	2.5	2	1.5	1
24	27	3	2	1.5	1
30	33	3.5	2	1.5	1
36	39	4	3	2	1.5
42	45	4.5	4	3	2
48	52	5	4	3	2
56	60	5.5	4	3	2
64	68	6	4	3	2
72	76	6	4	3	2
80	85	6	4	3	2
90	95	6	4	3	2
100	—	6	4	3	2
105	—	—	—	—	—
to	—	—	—	—	—
300	—	—	6	4	3

5.5 THREAD DESIGNATION

The diameter-pitch combination of an ISO metric screw thread is designated by the letter 'M' followed by the value of the nominal diameter and pitch, the two values being separated by the sign '×'. For example, a diameter pitch combination of nominal diameter 10 mm and pitch 1.25 mm is designated as M10 × 1.25.

If there is no indication of pitch in the designation, it shall mean the coarse pitch. For example, M 10 means that the nominal diameter of the thread is 10 mm and pitch is 1.5 mm.

Following are the other designations, depending on the shape of the thread profile :

SQ 40 × 10 – SQUARE thread of nominal diameter 40 mm and pitch 10 mm

ACME 40 × 8 – ACME thread of nominal diameter 40 mm and pitch 8 mm

WORM 40 × 10 – WORM thread of nominal diameter 40 mm and pitch 10 mm

5.8 REPRESENTATION OF THREADS

The true projection of a threaded portion of a part consists of a series of helices and it takes considerable time to draw them. Hence it is the usual practice to follow some conventional methods to represent screw threads. Figure 5.1 shows the true projection of a screw thread, whereas the conventional representation of external and internal threads as recommended by BIS is shown in Fig. 5.7.

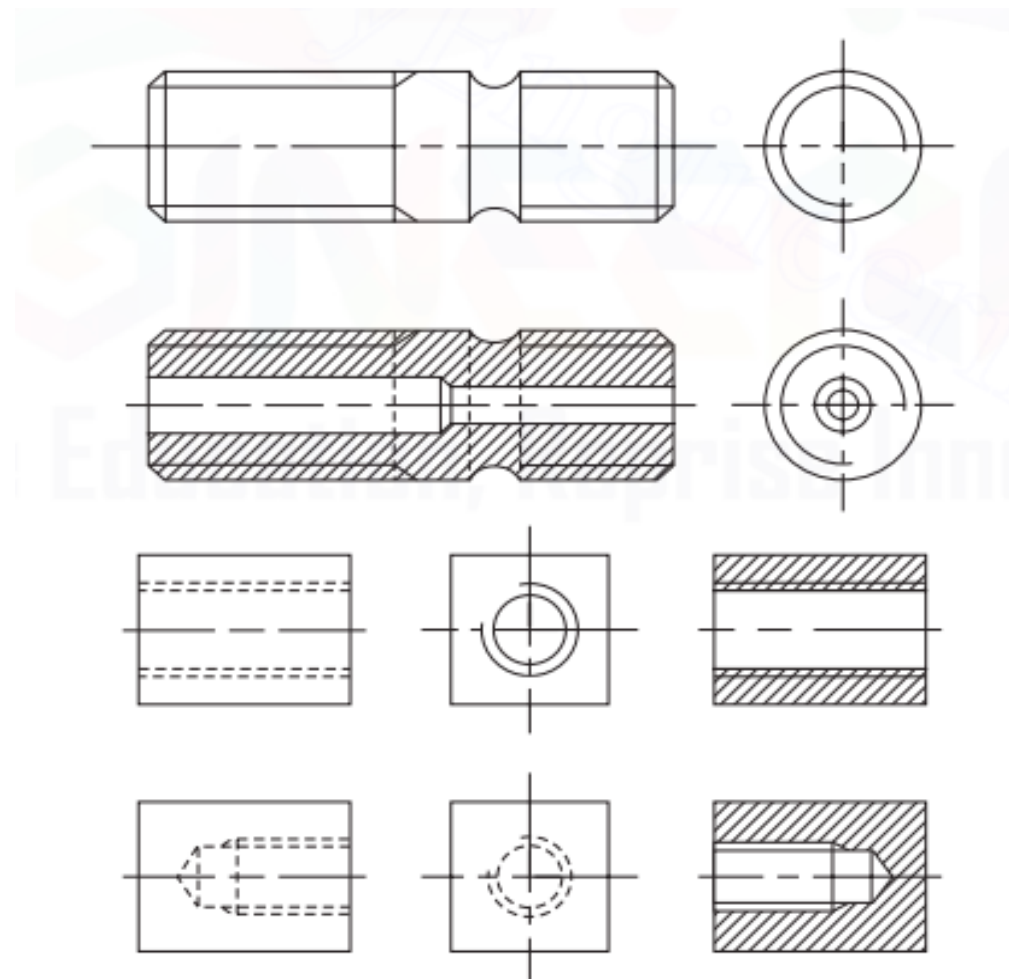


Fig. 5.7 Conventional representation of threads

5.8.1 Representation of Threaded Parts in Assembly

Figure 5.10 *a* shows the schematic representation and Figs. 5.10*b* and *c*, the conventional representation of threads in engagement.

Figure 5.10 *a* represents the internal threaded part in section; however, the external threaded one is shown unsectioned. In Figs. 5.10*b* and *c*, the external threaded parts are shown covering the internal threaded parts and should not be shown as hidden by them.

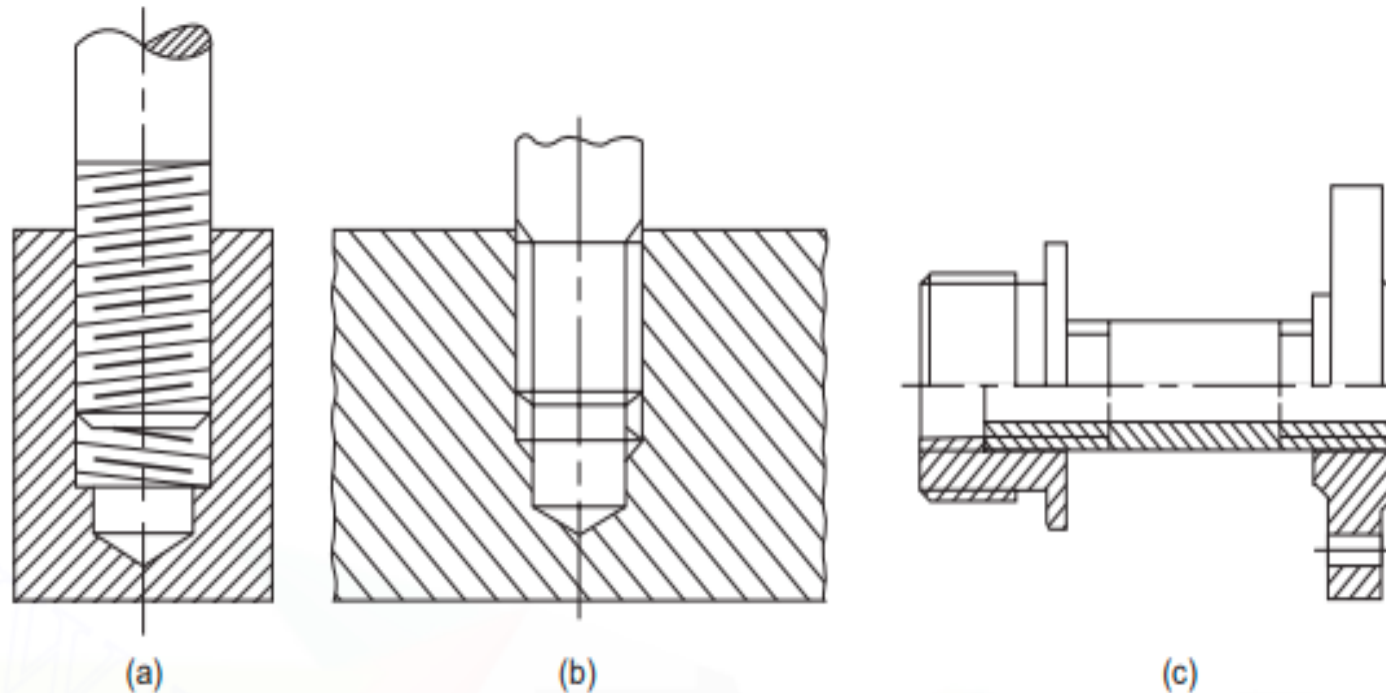


Fig. 5.10 External and internal threads in engagement

5.9 BOLTED JOINT

A bolt and nut in combination (Fig. 5.11) is a fastening device used to hold two parts together. The body of the bolt, called shank is cylindrical in form, the head; square or hexagonal in shape, is formed by forging. Screw threads are cut on the other end of the shank. Nuts in general are square or hexagonal in shape. The nuts with internal threads engage with the corresponding size of the external threads of the bolt. However, there are other forms of nuts used to suit specific requirements.

For nuts, hexagonal shape is preferred to the square one, as it is easy to tighten even in a limited space. This is because, with only one-sixth of a turn, the spanner can be re-introduced in the same position. However, square nuts are used when frequent loosening and tightening is required, for example on job holding devices like vices, tool posts in machines, etc.

The sharp corners on the head of bolts and nuts are removed by chamfering.

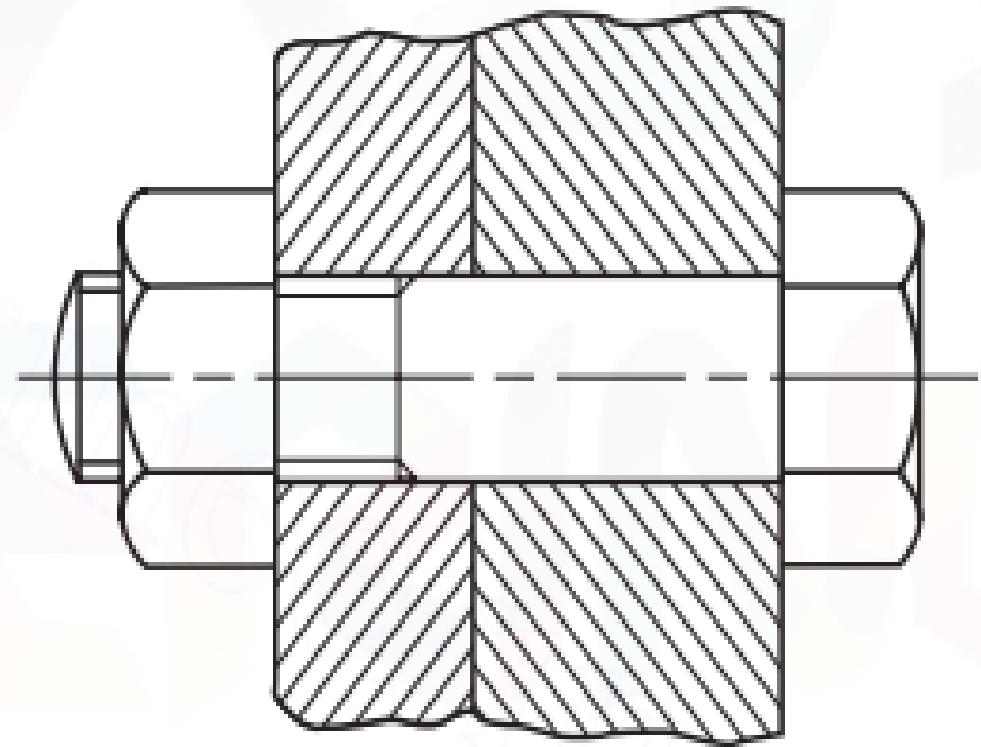


Fig. 5.11 Bolted joint

5.9.1 Methods of Drawing Hexagonal (Bolt Head) Nut

Drawing hexagonal bolt head or nut, to the exact dimensions is labourious and time consuming. Moreover, as standard bolts and nuts are used, it is not necessary to draw them accurately. The following approximate methods are used to save the draughting time :

Method 1 (Fig. 5.12)

Empirical relations :

Major or nominal diameter of bolt = D

Thickness of nut, T = D

Width of nut across flat surfaces, $W = 1.5D + 3 \text{ mm}$

Radius of chamfer, $R = 1.5D$

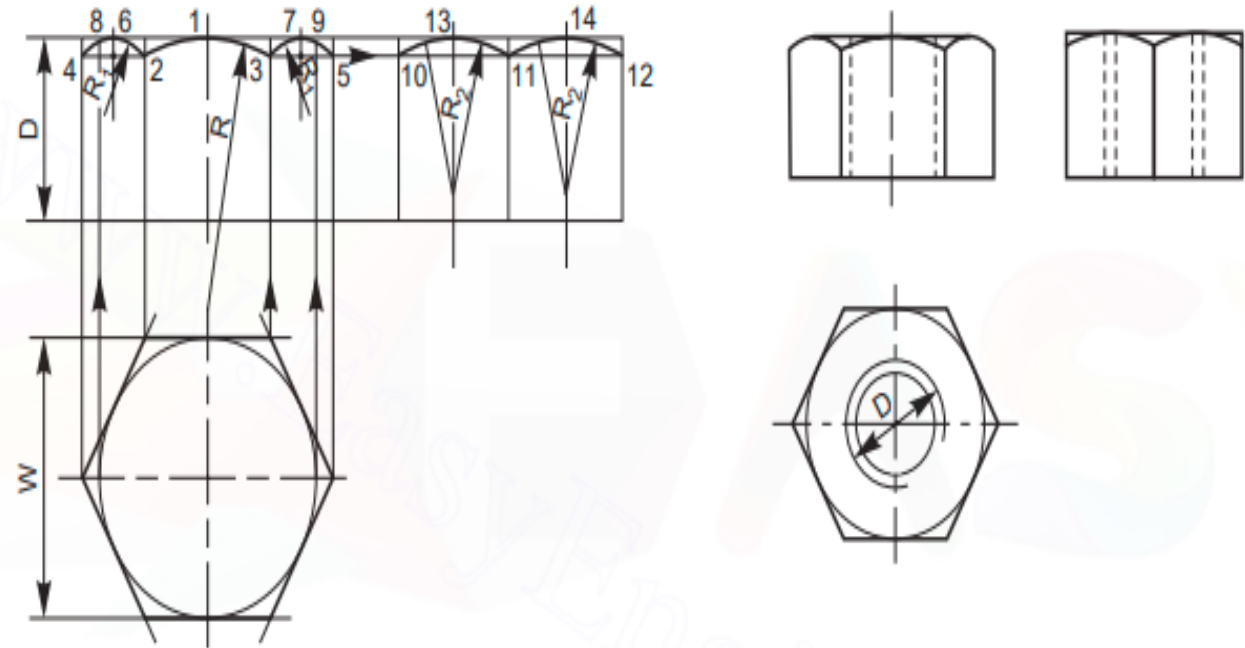
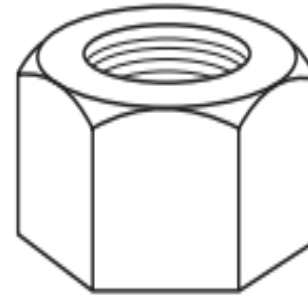


Fig. 5.12 Method of drawing views of a hexagonal nut (Method I)

Method 2 (Fig. 5.13)

Empirical relations :

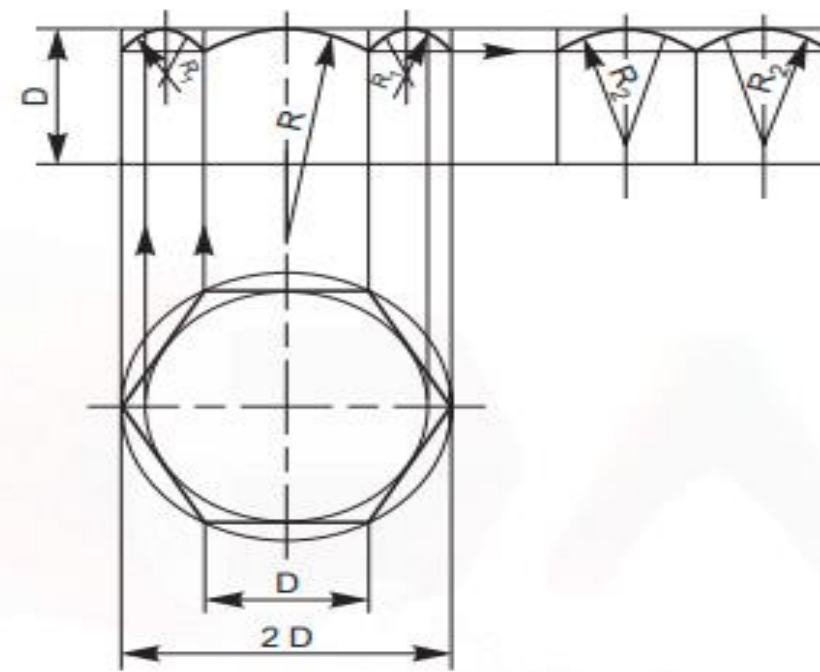
Major or nominal diameter of bolt = D

Thickness of nut, T = D

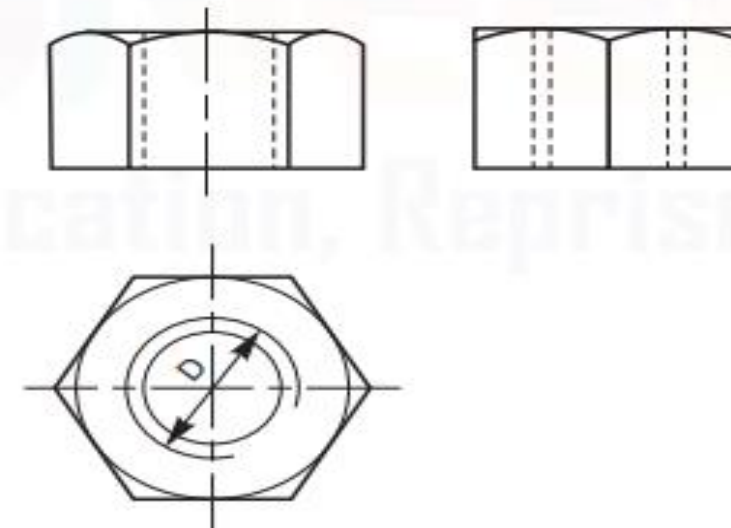
Width of the nut across corners = $2 D$

Radius of chamfer arc, R = $1.5 D$

Figure 5.13 illustrates the stages of drawing different views of a hexagonal nut, following the above relations, which are self-explanatory.



(a)



(b)

5.9.2 Method of Drawing Square (Bolt Head) Nut

A square bolt head and nut may be drawn, showing either across flats or corners. Following relations may be adopted for the purpose:

Major or nominal diameter of bolt	$= D$
Thickness of nut, T	$= D$
Width of the nut across flats, W	$= 1.5 D + 3 \text{ mm}$
Radius of chamfer arc, R	$= 2 D$

Figure 5.14 illustrates the method of drawing views of a square nut, in two orientations.

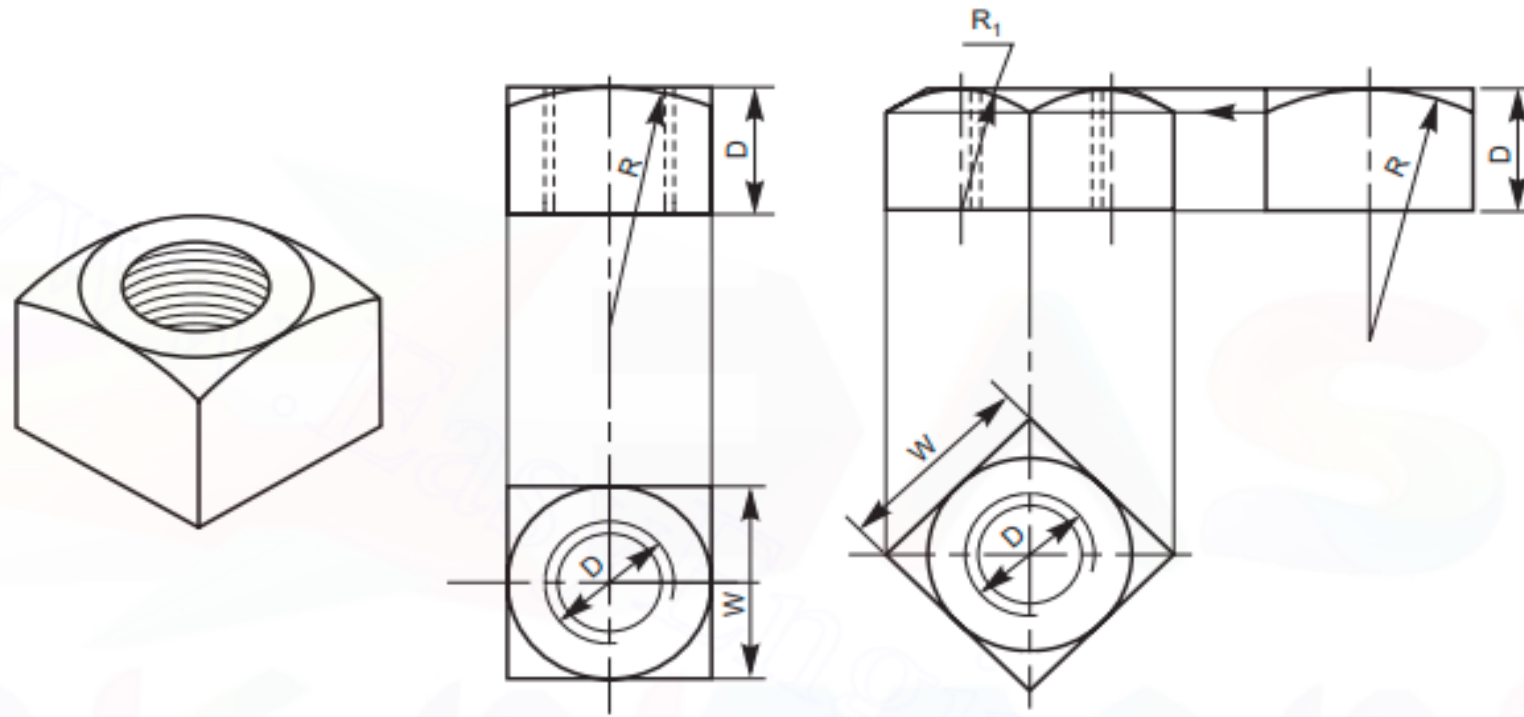


Fig.5.14 Method of drawing the views of a square nut

5.9.3 Hexagonal and Square Headed Bolts

Figure 5.15 shows the two views of a hexagonal headed bolt and square headed bolt, with the proportions marked.

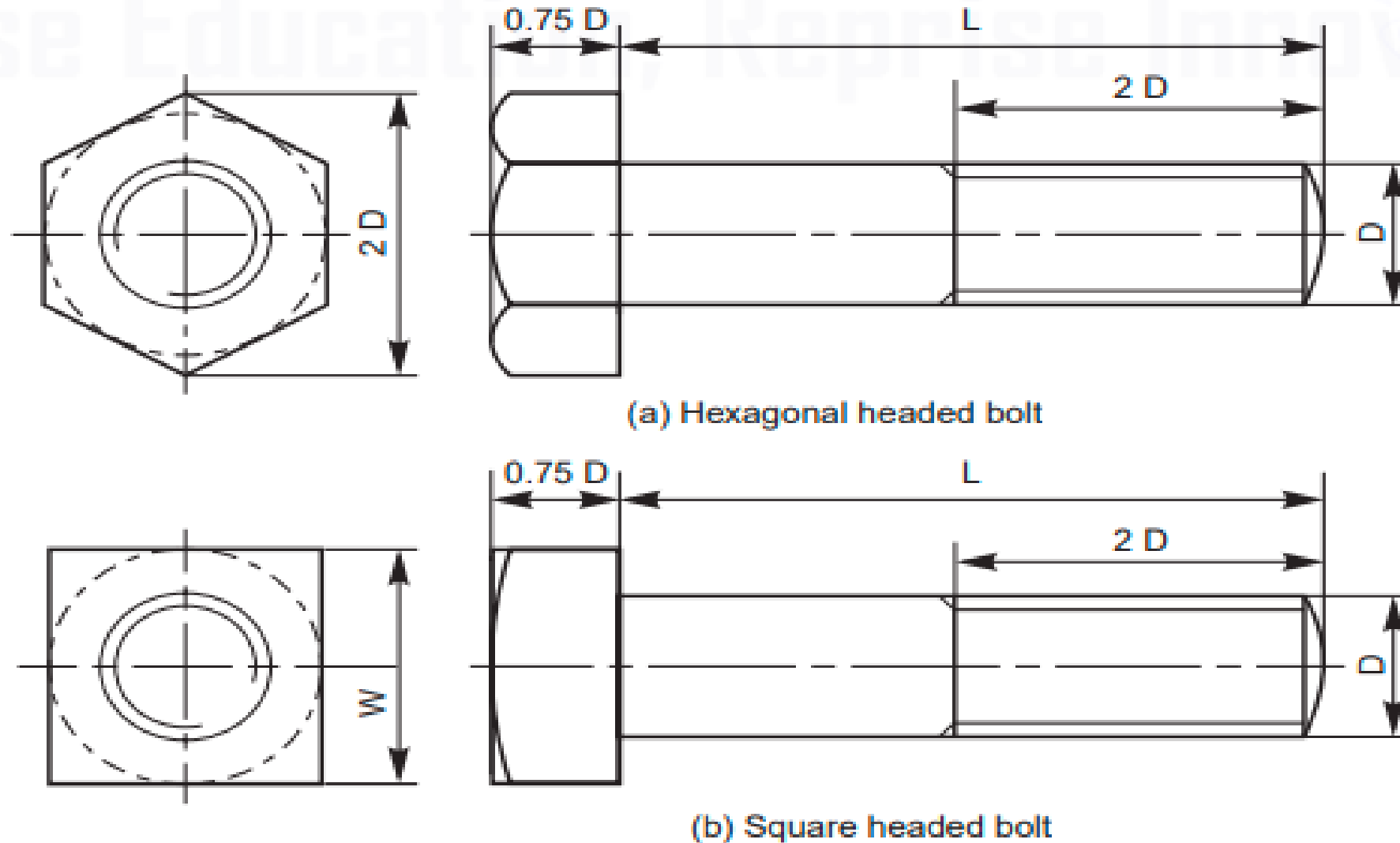


Fig. 5.15

5.9.4 Washers

A washer is a cylindrical piece of metal with a hole to receive the bolt. It is used to give a perfect seating for the nut and to distribute the tightening force uniformly to the parts under the joint. It also prevents the nut from damaging the metal surface under the joint. Figure 5.16 shows a washer, with the proportions marked.

Figure 5.17 illustrates the views of a hexagonal headed bolt with a nut and a washer in position.

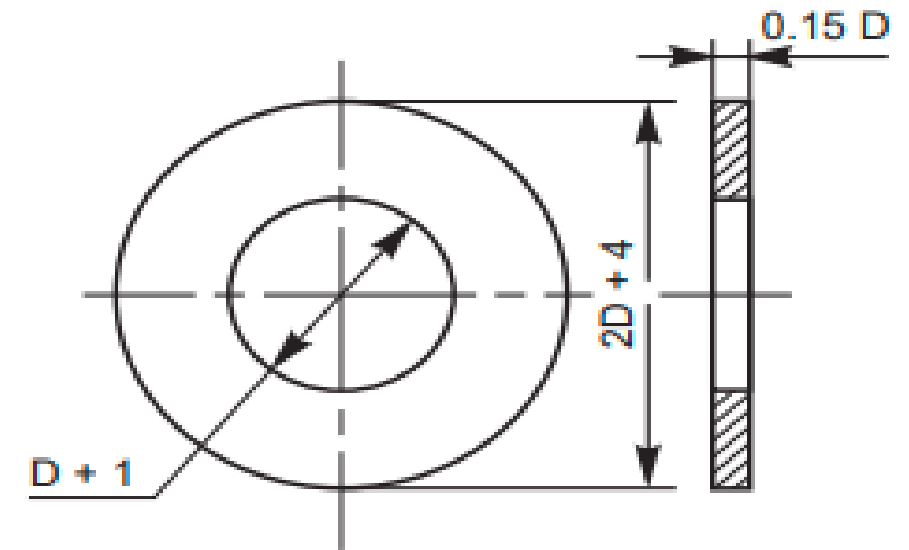


Fig. 5.16 Washer

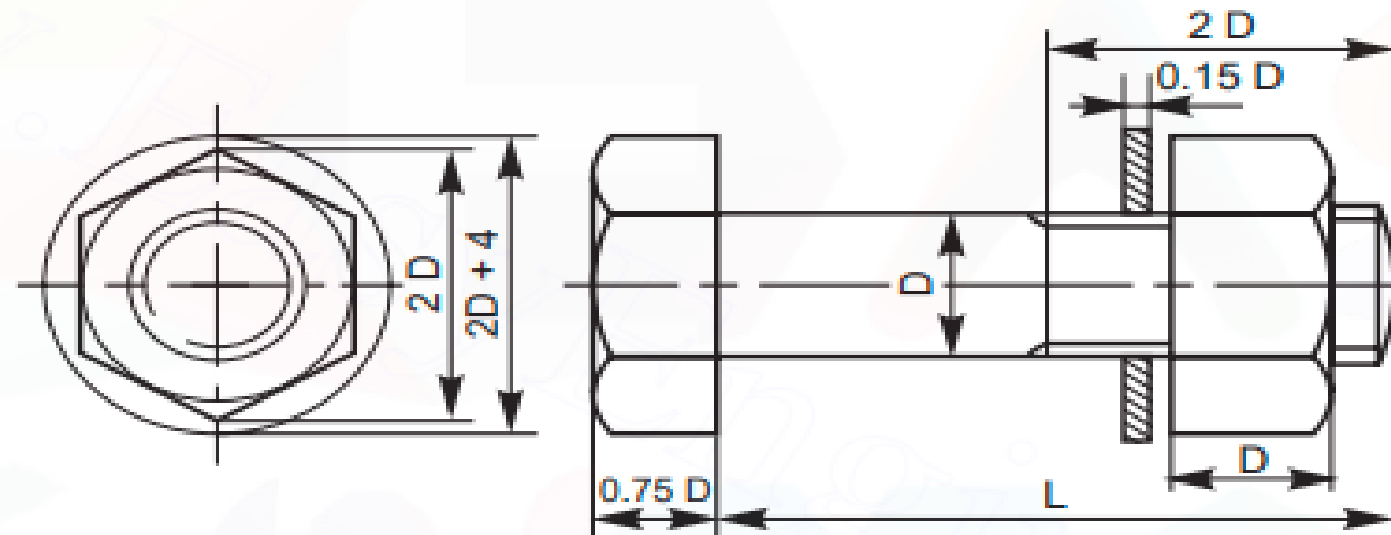


Fig. 5.17 A hexagonal headed bolt with a nut and a washer in position

Thank you!!!